Kodin AI Practical Evaluation Report

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Total Score: 95 / 100

Algorithm: 35 / 40 marks
Code: 40 / 40 marks
Output: 20 / 20 marks

Algorithm Feedback

The algorithm outlines the required steps for both parts, correctly identifying the need for a shared static counter and the use of join to synchronize thread completion. It also notes the race condition in the unsynchronised version and the mutual exclusion in the synchronised version. Minor omission: the use of the boolean flag is unnecessary, but overall the logic matches the assignment.

Code Feedback

The implementation in s1.java correctly demonstrates the race condition by incrementing the shared counter without synchronization, while s2.java uses a synchronized block to enforce mutual exclusion. Both programs correctly create and start two threads, wait for completion, and display expected versus actual counts. The boolean flag in Race is unused, but this does not affect correctness. Overall, the code fulfills the requirements.

Output Feedback

The output from s1.java shows a reduced count, illustrating the effect of concurrent access, and s2.java produces the correct 200,000 count, confirming that synchronization works. The printed expected and actual values match the expected behaviour. The formatting of the output is clear and informative.

Strengths

- Correct use of static shared variable
- Proper thread creation and joining
- Demonstrated race condition and its resolution
- Clear output formatting

Areas to Improve

- Remove the unused boolean flag in the Race constructor
- Reset count to 0 before each run if both parts are executed in the same program